

● MEDIUM

● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

07

10 questions · ~15 min

Q1

GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

The table shows the frequency of values in a data set.

Value	Frequency
19	7
21	1
23	7
25	4

What is the minimum value of the data set?

STUDENT-PRODUCED RESPONSE

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2

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3

3

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5

6

6

6

6

7

7

7

7

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8

9

9

9

9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

PROBLEM-SOLVING AND DATA ANALYSIS

Tanya earns \$13.50 per hour at her part-time job. When she works z hours, she earns $13.50z$ dollars. Which of the following expressions gives the amount, in dollars, Tanya will earn if she works $3z$ hours?

- A. $3(13.50z)$
- B. $3 + 13.50z$
- C. $3z + 13.50z$
- D. $13.50(z + 3)$

Q3

PROBLEM-SOLVING AND DATA ANALYSIS

From a population of 50,000 people, 1,000 were chosen at random and surveyed about a proposed piece of legislation. Based on the survey, it is estimated that 35% of people in the population support the legislation, with an associated margin of error of 3%. Based on these results, which of the following is a plausible value for the total number of people in the population who support the proposed legislation?

- A. 350
- B. 650
- C. 16,750
- D. 31,750

Q4

PROBLEM-SOLVING AND DATA ANALYSIS

The number k is 36% greater than 50. If k is the product of 50 and r , what is the value of r ?

- A. 36
- B. 3.6
- C. 1.36
- D. 0.36

Q5

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

At a conference, there are a total of 275 attendees. Each attendee is assigned to either group A, group B, or group C. If one of these attendees is selected at random, the probability of selecting an attendee who is assigned to group A is 0.44 and the probability of selecting an attendee who is assigned to group B is 0.24. How many attendees are assigned to group C?

STUDENT-PRODUCED RESPONSE

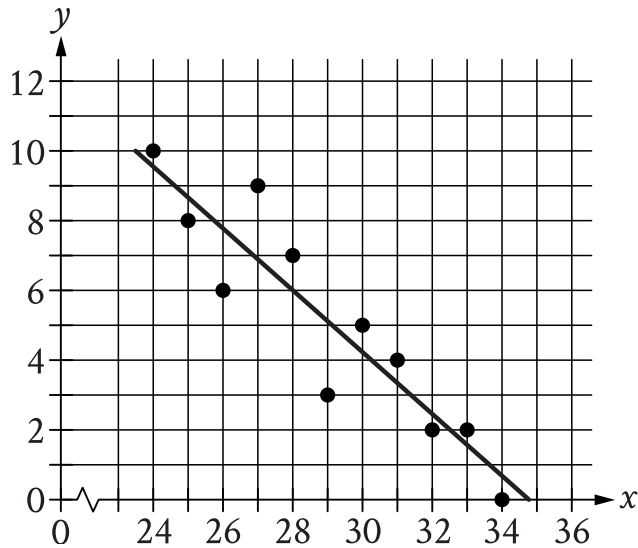
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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The scatterplot shows the relationship between two variables, x and y . A line of best fit for the data is also shown.



- The scatterplot has 11 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
 - The line of best fit slants down from left to right.
 - 6 points are above the line of best fit.
 - 5 points are below the line of best fit.
 - The line of best fit goes through the following approximate coordinates:
 - (28 comma 6)
 - (33 comma 1.5)

At $x = 32$, which of the following is closest to the y -value predicted by the line of best fit?

A. 0.4

B. 1.5

C. 2.4

D. 3.3

Q7

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

If a is the mean and b is the median of nine consecutive integers, what is the value of $|a - b|$?

STUDENT-PRODUCED RESPONSE

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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

The ratio x to y is equivalent to the ratio 9 to 5. If the value of x is 162, what is the value of y ?

STUDENT-PRODUCED RESPONSE

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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

PROBLEM-SOLVING AND DATA ANALYSIS

A bag containing 10,000 beads of assorted colors is purchased from a craft store. To estimate the percent of red beads in the bag, a sample of beads is selected at random. The percent of red beads in the bag was estimated to be 15%, with an associated margin of error of 2%. If r is the actual number of red beads in the bag, which of the following is most plausible?

- A. $r > 1,700$
- B. $1,300 < r < 1,700$
- C. $200 < r < 1,500$
- D. $r < 1,300$

Q10

PROBLEM-SOLVING AND DATA ANALYSIS

In March, the price of a collectible card was \$ 15.50. In April, the price of the collectible card was \$ 17.36. The price of the collectible card in April was $p\%$ of the price of the collectible card in March. What is the value of p ?

- A. 12
- B. 88
- C. 112
- D. 188